Proofs & Counterexamples

1. Prove or give a counterexample to each of the following "existential statements".

(a) There exists a prime number p with $90 < p < 105$.

(b) There exists a real number r with $r^2 + 2r + 2 \leq 0$.

(c) There exists a sequence of 100 consecutive composite numbers.

2. Prove or give a counterexample to each of the following "universal statements".

(a) For every real number r, $r^2 + 2r + 1 \geq 0$.

(b) For all positive integers n, the number $\frac{n(n+1)}{2}$ is an integer.

(c) Every prime number is odd.

Prove or give a counterexample to each of the following statements.

3. Let a and b be integers. If $a \,\big|\, b$, then $a \,\big|\, 3b^3 - 4b^2 + 18b + 2a$.

4. Let m and n be integers. If m and n have different parity, then their sum is odd.

5. Let x be a real number. If $x^2 + 3x < 0$, then $x < 0$.

6. (Careful!) Let a and b be integers. If $a^2(b^2 - 2b)$ is odd, then a and b must both be odd.